

决策树

Decision Tree

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声明

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定义

Definition

- ▶ 决策树是一种以树状结构来表示选择及其结果的图。其中，节点表示事件或选择，边表示决策规则或条件。顾名思义，决策树学习算法会构建一个树状结构的模型。
 - ▶ 该模型本身由一系列逻辑判断组成，类似于流程图；其中的决策节点表示需要基于某个特征属性做出的判断。
 - ▶ 这些节点会延伸出不同的分支，代表该判断对应的各类选择。
 - ▶ 树的末端由叶节点（也称为终端节点）收尾，代表经过一系列判断组合后得出的最终结果。
- ▶ Decision tree is a graph to represent choices and their results in form of a tree. The nodes in the graph represent an event or choice and the edges of the graph represent the decision rules or conditions. As intuited from the name, decision tree builds a model in the form of a tree structure.
 - ▶ The model itself comprises a series of logical decisions, similar to a flowchart, with decision nodes that indicate a decision to be made on an attribute.
 - ▶ These split into branches that indicate the decision's choices.
 - ▶ The tree is terminated by leaf nodes (also known as terminal nodes) that denote the result of following a combination of decisions.

应用

Application

- ▶ 决策树在实际中有较多的应用，下面列举了一些常见的应用实例：
 - ▶ 判断一封邮件是否为垃圾邮件；
 - ▶ 基于医学检测数据、症状或疾病进展速率进行的病症诊断。例如，判断肿瘤是否为恶性；
 - ▶ 根据各项指标评估一笔贷款的信贷风险是优质还是不良。注意在信用评分模型中，导致申请者被拒绝的各项判定准则需要被清晰、明确地定义。
- ▶ Decision tree has a variety of applications in practice, some are enumerated as in the below:
 - ▶ Predicting an email as spam or not spam;
 - ▶ Diagnosis of medical conditions based on laboratory measurements, symptoms, or rate of disease progression. For example, predicting of a tumor is cancerous;
 - ▶ Predicting a loan as a good or bad credit risk based on the factors in each of these. Notably, Credit scoring models in which the criteria that causes an applicant to be rejected need to be well-specified.

分治策略

Divide and Conquer

- ▶ 决策树通过称为递归划分的启发式方法构建，这种方法通常被称为分治法，其利用特征值，将数据不断分割成类别相似度更高的子集。具体地，从代表整个数据集的根节点开始，算法选择一个对目标类别预测能力最强的特征，随后，算法根据该特征的不同取值将样本划分成不同组别，这一判断形成了树的第一层分支。

Decision trees are built using a heuristic called recursive partitioning. This approach is generally known as divide and conquer because it uses the feature values to split the data into smaller and smaller subsets of similar classes. Beginning at the root node, which represents the entire dataset, the algorithm chooses a feature that is the most predictive of the target class. The examples are then partitioned into groups of distinct values of this feature; this decision forms the first set of tree branches.

- ▶ 根据特征变量的选择与节点分支的策略，重复上面过程，生成最终的决策树。

Based on the strategy for selecting feature variables and branching nodes, repeat the above process to generate the final decision tree.

示例

Example

- 为了说明决策树的构建过程，我们来看一个简单的例子。假设我们在某电影制片厂工作，需要从待审的剧本中选出合适的剧本进行拍摄。为减少工作量，避免逐字逐句地阅读每一个剧本，需要开发一个决策树算法来预测一部潜在电影的类别。这些类别包括：主流热门影片、影评人青睐之作或票房失败之作。

To illustrate the tree building process, let's consider a simple example. Imagine that we are working for some film studio, and to select from a pile of screenplays for future shooting. To reduce the workload, aka., avoiding reading each one cover-to-cover, we decide to develop a decision tree algorithm to predict whether a potential movie would fall into one of three categories: mainstream hit, critic's choice, or box office bust.

- 为了给模型收集数据，我们查阅了制片厂的档案，分析了过去十年上映的电影。在审阅了30个不同电影剧本的数据后，一种规律开始显现。即影片的拟定拍摄预算、领衔主演的一线明星数量，与影片的成功类别之间存在着关联。

To gather data for the model, we examined the previous ten years of movie releases in the studio archives. After reviewing the data for 30 different movie scripts, a pattern emerges. Specifically, there exists a relationship between the film's proposed shooting budget, the number of A-list celebrities lined up for starring roles, and the categories of success.

示例

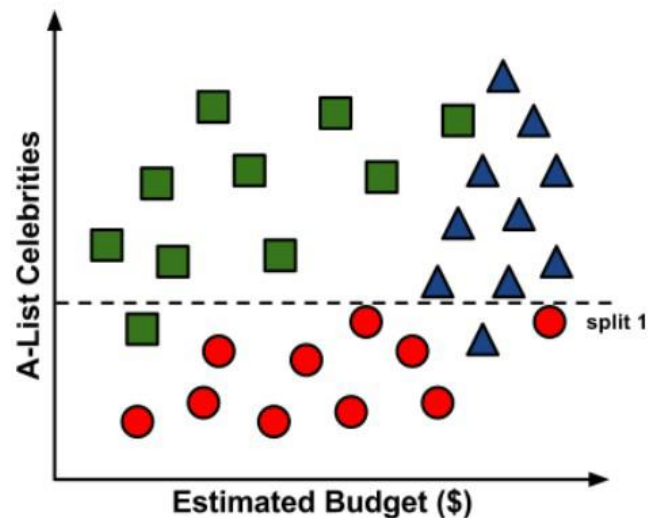
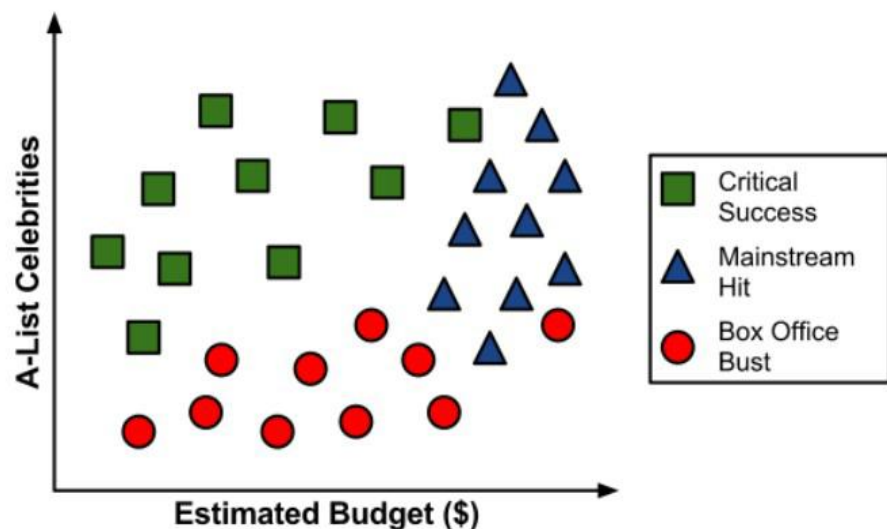
Example

- 我们先看影片结果与预算、参演一线明星数量之间的关系，其散点图如左下所示。

We first examine the box office versus the budget and participated number of A-list celebrities. A scatter plot of this data are in the below.

- 我们根据参演明星数量，对散点图进行划分。

We partition the scatter plot based on the number of starring celebrities.



示例

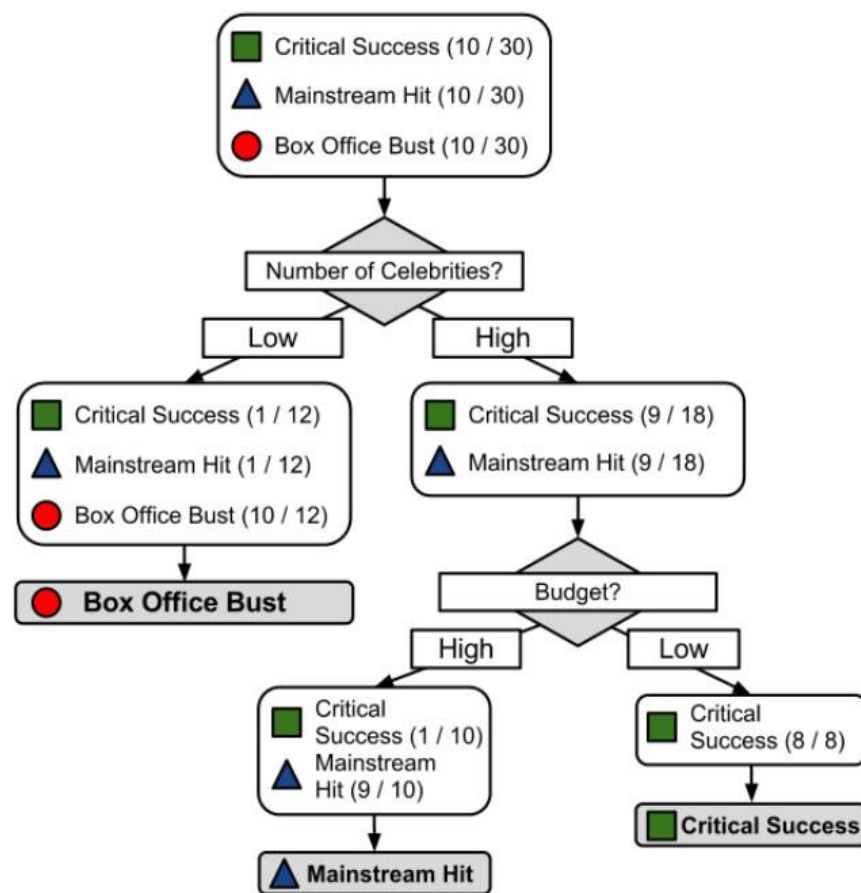
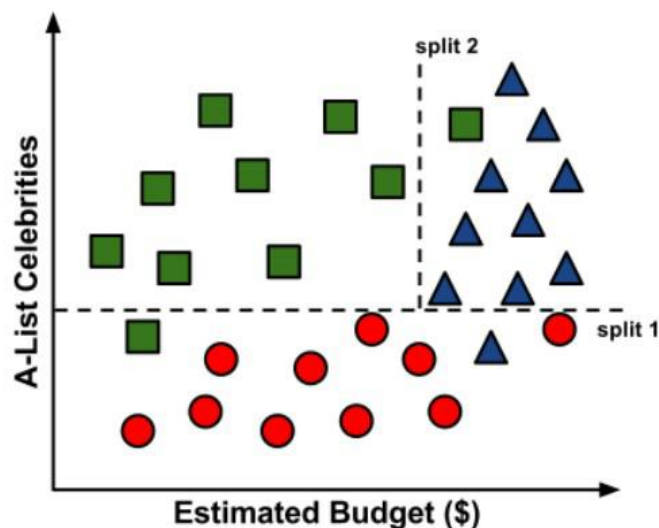
Example

- ▶ 然后我们根据预算，对散点图进行进一步划分。

We partition the scatter plot based on the budget for further partition.

- ▶ 此时，我们基本得到了对应的三个类别，构建的相应的决策树如右所示：

Now, we basically reach the three categories, and the constructed decision tree accordingly is illustrated as in the right:



准则

Criteria

- ▶ 从上述过程可以看到，我们构建树的过程一般不唯一。那么如何评判一棵优质的决策树？

与训练数据保持一致

- ▶ 能够对训练样本进行正确分类

对未知数据表现良好

- ▶ 能够对未来的新输入数据进行正确分类

结构尽可能精简

- ▶ 实现高效的分类运算

- ▶ 但问题是如何找到好的决策树呢？由于可能生成的决策树数量极其庞大，因此无法通过穷举法实现。我们下面介绍一些常用的算法。

- ▶ From the above procedure, it can be concluded that there are multiple way led to different decision trees being constructed. So, What is a Good Decision Tree?

Consistent with training data

- ▶ classifies training examples correctly

Performs well on future examples

- ▶ classifies future inputs correctly

As small as possible

- ▶ Efficient classification

- ▶ The problem remaining is how can we find a small decision tree? Because there are possible decision trees, so brute force is not possible. We will introduce some method to construct decision tree.

ID3算法

ID3 Algorithm

- 下面讲解构建决策树常用的ID3算法，其由学者Ross Quinlan发明。由ID3算法构建的决策树，其节点分为中间节点（属性节点）或叶节点（分类节点），边代表对应属性的值。决策树整体上应尽可能简单。

Now we introduce the simple but commonly-used ID3 algorithm invented by Ross Quinlan for constructing decision tree. Under this algorithm, each of the tree nodes can either be an intermediate/interior node (attribute node) or a leaf node (classification node), with outgoing edges labelled with attribute values. Grossly, tree should be as small as possible.

- 从定性描述的角度，在基于训练数据从根节点开始构建的过程中，选择能将数据划分为若干子集的、能实现“良好划分”的属性。所谓良好划分，指子节点对应的子集属于同一类别，这样可以直接得出分类结果。若子节点对应的子集包含不同类别，则通过递归方式继续处理。

From a qualitative descriptive perspective, start at root with the training data, choose attribute that creates a “good split”, aka., children with subsets that are within the same classification, which lead to a classification. For children with subsets that are mixed, continue handling them with recursion.

熵

Entropy

- 下面从定量的角度描述特征选择。在这之前，首先需要介绍信息学中熵的概念。那么，什么是信息？信息是衡量不确定性的量。考虑事件 A ，其发生具有随机性，则其信息量定义为 $I(A) = -\log_2 P(A)$ ，即其发生的概率的对数的取反值。

Now we describe the attribute selection from a qualitative perspective. Before that, we introduce the concept of entropy in the information theory. First, what is information? Information is the quantity for measuring uncertainty. For example, considering a random event A , the information it contains is defined as $I(A) = -\log_2 P(A)$, aka, the negative of the logarithm of its probability.

- 以掷硬币为例，记 A 为出现正面的概率。若硬币无瑕，则出现正面的概率为0.5，此时 $I_{\text{perfect}}(A) = \log_2 2 = 1$ ；若硬币有瑕，导致出现正面概率为0.25，则 $I_{\text{defect}}(A) = 2 \log_2 2 = 2$ ，即此种情况下出现正面更会让人惊讶。

Take tossing a coin for example, let A denote the case of head up. For a prefect coin, we know $P(A) = 0.5$, which induces an information quantity $I_{\text{perfect}}(A) = \log_2 2 = 1$. A defect coin might reduce the chance of head up from 0.5 to 0.25, now we have $I_{\text{defect}}(A) = 2 \log_2 2 = 2$, which means when it happens, it surprise us more.

熵

Entropy

- ▶ 假设一个随机试验 S 有 N 种结果 $\{x_1, x_2, \dots, x_N\}$, 每种结果的信息量的加权平均值定义为 S 的熵, 即 $H(S) = -\sum_{i=1}^N P(x_i) \log_2 P(x_i)$ 。

Assume the outcomes of an experiment are enumerated as $\{x_1, x_2, \dots, x_N\}$, the weighted sum of the information of each outcome is called the entropy of S , defined as $H(S) = -\sum_{i=1}^N P(x_i) \log_2 P(x_i)$.

- ▶ 以掷硬币为例, 若硬币无瑕, 出现正面与反面的概率均为0.5, 此时

Take tossing a coin for example, let A denote the case of head up. For a perfect coin, we know $P(A) = 0.5$, which induces an information quantity

$$H_{\text{perfect}}(S) = 0.5 * \log_2 2 + 0.5 * \log_2 2 = 1$$

- ▶ 若硬币有瑕, 导致出现正面概率为0.25, 反面概率为0.75, 则

A defect coin might reduce the chance of head up from 0.5 to 0.25, now we have

$$H_{\text{defect}}(S) = -0.25 * \log_2 0.25 - 0.75 \log_2 0.75 = 0.8113$$

- ▶ 由于 $H_{\text{perfect}}(S) > H_{\text{defect}}(S)$, 熵减小, 即表明此时情况更容易猜结果。

Since $H_{\text{perfect}}(S) > H_{\text{defect}}(S)$, decrease in entropy means more easier to guess the outcome.

信息增益

Information Gain

- 接下来，我们介绍信息增益的概念。信息增益表示在知道某个特征之后使得的不确定性减少的程度(知道某个特征前的熵与知道某个特征之后的熵之差)。根据某个变量将样本数据分割为多个子集，分割前与分割后样本数据的熵之差为信息增益，信息增益越高，表示该变量对样本数据的分类效果越好。

Next, we introduce the concept of information gain. Information gain indicates the degree to which uncertainty is reduced after knowing a certain feature (it is the difference between the entropy before knowing the feature and the entropy after knowing the feature). When sample data is divided into multiple subsets based on a variable, the difference in entropy of the sample data before and after the division is the information gain. The higher the information gain, the better the classification effect of that variable on the sample data.

- 用数学语言对信息增益进行描述：特征 A 对训练数据集 $\mathcal{D}$ 的信息增益 $g(\mathcal{D}, A)$ ，定义为集合 $\mathcal{D}$ 经验熵 $H(\mathcal{D})$ 与特征 A 给定条件下 $\mathcal{D}$ 的经验条件熵 $H(\mathcal{D}|A)$ 之差。

We describe information gain in mathematical terms: The information gain $g(\mathcal{D}, A)$ of feature A with respect to the training dataset $\mathcal{D}$ is defined as the difference between the empirical entropy $H(\mathcal{D})$ of set $\mathcal{D}$ and the empirical conditional entropy $H(\mathcal{D}|A)$ of $\mathcal{D}$ given feature A .

信息增益

Information Gain

- 具体地，假设训练数据集为 $\mathcal{D}$ ， $|\mathcal{D}|$ 表示数据集中样本的个数。假设数据 $\mathcal{D}$ 中有 K 个类 C_k ($k = 1, 2, \dots, K$)， $|C_k|$ 表示属于类 C_k 的样本个数，我们有 $\sum_{k=1}^K |C_k| = |\mathcal{D}|$ 。进一步地，假设 A 为数据集 $\mathcal{D}$ 中的某一个特征变量，该特征变量有 n 个不同的取值 $\{a_1, a_2, \dots, a_n\}$ 。根据特征 A 的取值，将 $\mathcal{D}$ 划分为 n 个子集 $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_n$ 。令 $|\mathcal{D}_i|$ 表示为子集 $\mathcal{D}_i$ 的样本个数，则有 $\sum_{i=1}^n |\mathcal{D}_i| = |\mathcal{D}|$ 。记子集 $\mathcal{D}_i$ 中属于类 C_k 的样本集合为 $\mathcal{D}_{ik}$ ，则 $|\mathcal{D}_{ik}|$ 为 $\mathcal{D}_{ik}$ 的样本个数。基于此假定，信息增益计算如右所示：

Specifically, assume the training dataset is $\mathcal{D}$, and $|\mathcal{D}|$ denotes the number of samples in the dataset. Assume there are K classes C_k ($k = 1, 2, \dots, K$) in dataset $\mathcal{D}$, where $|C_k|$ represents the number of samples belonging to class C_k . We have $\sum_{k=1}^K |C_k| = |\mathcal{D}|$. Furthermore, assume A is a feature variable in dataset $\mathcal{D}$ that takes n distinct values $\{a_1, a_2, \dots, a_n\}$. According to the values of feature A , dataset $\mathcal{D}$ is divided into n subsets $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_n$. Let $|\mathcal{D}_i|$ denote the number of samples in subset $\mathcal{D}_i$; then we have $\sum_{i=1}^n |\mathcal{D}_i| = |\mathcal{D}|$. Denote the set of samples in subset $\mathcal{D}_i$ that belong to class C_k as $\mathcal{D}_{ik}$, then $|\mathcal{D}_{ik}|$ is the number of samples in $\mathcal{D}_{ik}$. Based on this assumption, the information gain can be calculated in the right:

$$g(\mathcal{D}, A) = H(\mathcal{D}) - H(\mathcal{D}|A)$$

$$H(\mathcal{D}) = - \sum_{k=1}^K \frac{|C_k|}{|\mathcal{D}|} \log_2 \frac{|C_k|}{|\mathcal{D}|}$$

$$\begin{aligned} H(\mathcal{D}|A) &= \sum_{i=1}^n \frac{|\mathcal{D}_i|}{|\mathcal{D}|} H(\mathcal{D}_i) \\ &= \sum_{i=1}^n \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \left(- \sum_{k=1}^K \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \right) \\ &= - \sum_{i=1}^n \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \sum_{k=1}^K \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \end{aligned}$$

ID3算法

ID3 Algorithm

- 现在，我们以一个例子，通过计算信息增益，来说明如何选择当前最合适的属性来分裂节点。右表是一个根据天气情况确定是否适合户外运动的例子。我们希望构建一个决策树，当有新数据时，判断是否适合进行户外运动。

We now use an example to illustrate how to select the most suitable attribute for splitting the current node by calculating information gain. The right table shows decision on outdoor activities based on weather conditions. We aim to construct a decision tree to judge the suitability of outdoor activities when new data becomes available.

Day	Outlook (feature)	Temp. (feature)	Humidity (feature)	Wind (feature)	Decision (outcome)
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

ID3算法

ID3 Algorithm

- 我们首先来决定根节点的属性，根据上表，我们可以从展望、温度、湿度、风力四个元素里面选。我们先计算各自的信息增益。

We first need to decide the attribute for the root node. According to the table above, we can choose from outlook, temperature, humidity and wind. We first calculate the information gain of each attribute respectively.

- 我们首先根据公式计算 $H(\mathcal{D})$ ，如下所示。注意到，最后决策分为两类，进行户外活动或不进行。我们简单地用频率计算先验概率：

First we calculate the parent entropy, aka., before classification using that attribute, as in the following. Note that the final decision can be categorized into two case, YES or NO. We using the frequency of each case to approach the prior probability:

$$H(\mathcal{D}) = - \sum_{k=1}^K \frac{|C_k|}{|\mathcal{D}|} \log_2 \frac{|C_k|}{|\mathcal{D}|} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

ID3算法

ID3 Algorithm

- 下面我们来计算关于展望的 $H(\mathcal{D}|A)$ 。为方便起见，我们先根据展望构造表如下：

Now we calculate $H(\mathcal{D}|A)$ for outlook. For convenience, we first construct the following table:

Outlook	Yes	No	#instance
Sunny	$2(P_{\text{yes}} = 2/5)$	$3(P_{\text{no}} = 3/5)$	5
Overcast	$4(P_{\text{yes}} = 4/4)$	$0(P_{\text{no}} = 0/4)$	4
Rain	$3(P_{\text{yes}} = 3/5)$	$2(P_{\text{no}} = 2/5)$	5

$$\begin{aligned} H(\mathcal{D}|A) &= \sum_{i=1}^3 \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \sum_{k=1}^2 -\frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \\ &= \frac{5}{14} \left[-\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \right] + \frac{4}{14} \left[-\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} \right] + \frac{5}{14} \left[-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right] = 0.69 \end{aligned}$$

ID3算法

ID3 Algorithm

- 同理，关于温度的 $H(\mathcal{D}|A)$ 计算如下：

Similarly, we calculate $H(\mathcal{D}|A)$ for temperature in the below:

Temperature	Yes	No	#instance
Hot	$2(P_{\text{yes}} = 2/4)$	$2(P_{\text{no}} = 2/4)$	4
Mild	$4(P_{\text{yes}} = 4/6)$	$2(P_{\text{no}} = 2/6)$	6
Cool	$3(P_{\text{yes}} = 3/4)$	$1(P_{\text{no}} = 1/4)$	4

$$\begin{aligned} H(\mathcal{D}|A) &= \sum_{i=1}^3 \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \sum_{k=1}^2 -\frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \\ &= \frac{4}{14} \left[-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \right] + \frac{6}{14} \left[-\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} \right] + \frac{4}{14} \left[-\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right] = 0.91 \end{aligned}$$

ID3算法

ID3 Algorithm

- 同理，关于湿度的 $H(\mathcal{D}|A)$ 计算如下：

Similarly, we calculate $H(\mathcal{D}|A)$ for humidity in the below:

Outlook	Yes	No	#instance
High	$3(P_{\text{yes}} = 3/7)$	$4(P_{\text{no}} = 4/7)$	7
Normal	$6(P_{\text{yes}} = 6/7)$	$1(P_{\text{no}} = 1/7)$	7

$$\begin{aligned} H(\mathcal{D}|A) &= \sum_{i=1}^2 \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \sum_{k=1}^2 -\frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \\ &= \frac{7}{14} \left[-\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} \right] + \frac{7}{14} \left[-\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} \right] = 0.79 \end{aligned}$$

ID3算法

ID3 Algorithm

- 同理，关于风力的 $H(\mathcal{D}|A)$ 计算如下：

Similarly, we calculate $H(\mathcal{D}|A)$ for wind in the below:

Temperature	Yes	No	#instance
Weak	$6(P_{\text{yes}} = 6/8)$	$2(P_{\text{no}} = 2/8)$	8
Strong	$3(P_{\text{yes}} = 3/6)$	$3(P_{\text{no}} = 3/6)$	6

$$\begin{aligned} H(\mathcal{D}|A) &= \sum_{i=1}^2 \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \sum_{k=1}^2 -\frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \\ &= \frac{8}{14} \left[-\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} \right] + \frac{6}{14} \left[-\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} \right] = 0.89 \end{aligned}$$

ID3算法

ID3 Algorithm

- 根据上面结果，我们计算相应的信息增量，如下表所示：

Now we calculate the information gain according to the table below:

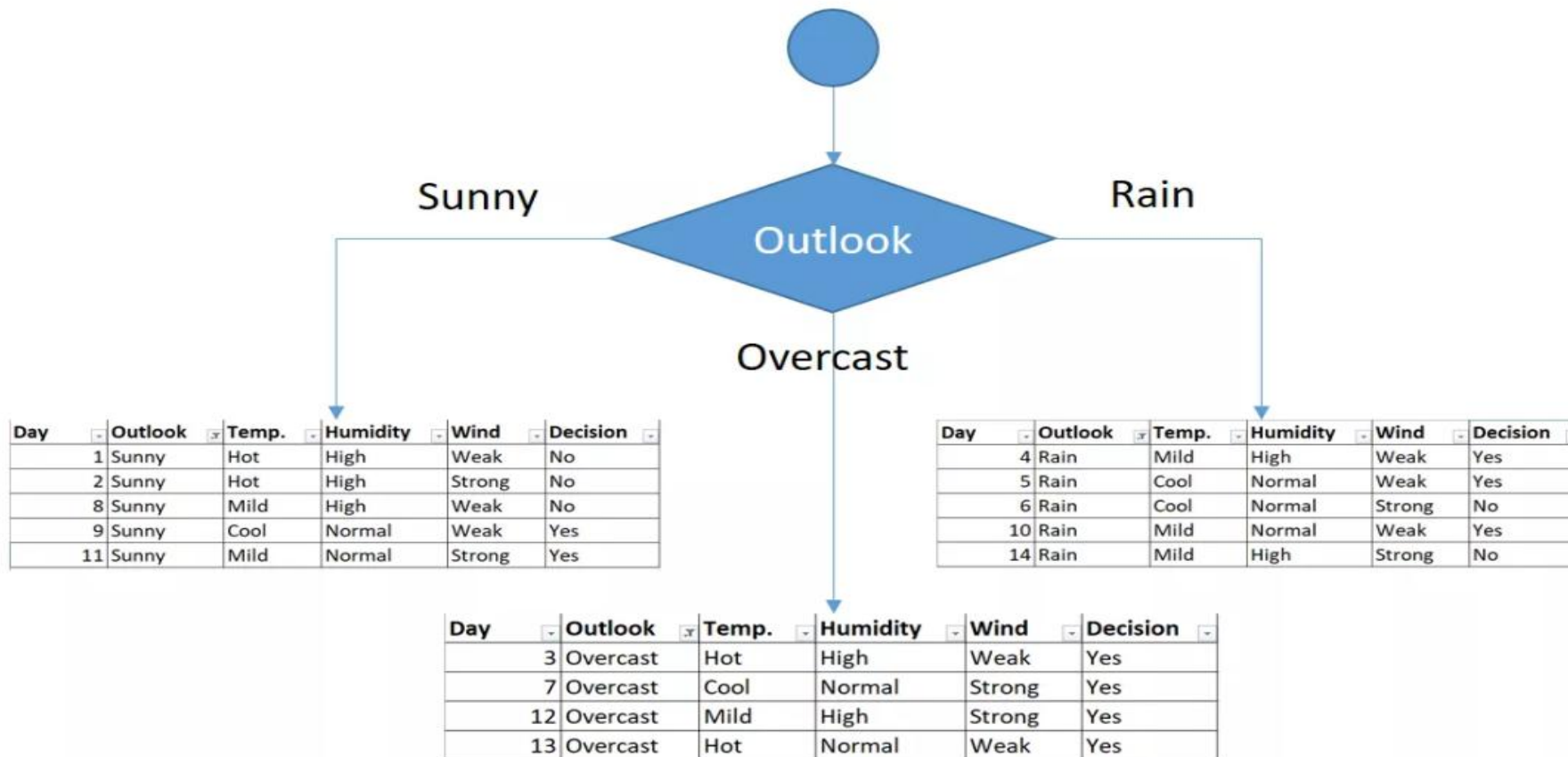
$H(\mathcal{D})$	$H(\mathcal{D} \text{Outlook})$	$H(\mathcal{D} \text{Temp.})$	$H(\mathcal{D} \text{Humidity})$	$H(\mathcal{D} \text{Wind})$	Choice
0.94	0.69	0.91	0.79	0.89	
$g(\mathcal{D}, A)$	0.25	0.03	0.15	0.05	Outlook

- 从表中可以看出，根据上面结果，关于属性或特征outlook的信息增量最大。ID3算法中，是选择信息增量最大的属性作为节点，往下建树。所建的第一层级的树如下页所示：

From the table that, it is obvious that the attribute/feature “outlook” has the largest information gain based on the previous results. In the ID3 algorithm, the attribute with the largest information gain is selected as the node to grow the decision tree downward. The first-level tree constructed is shown on the next page:

ID3算法

ID3 Algorithm

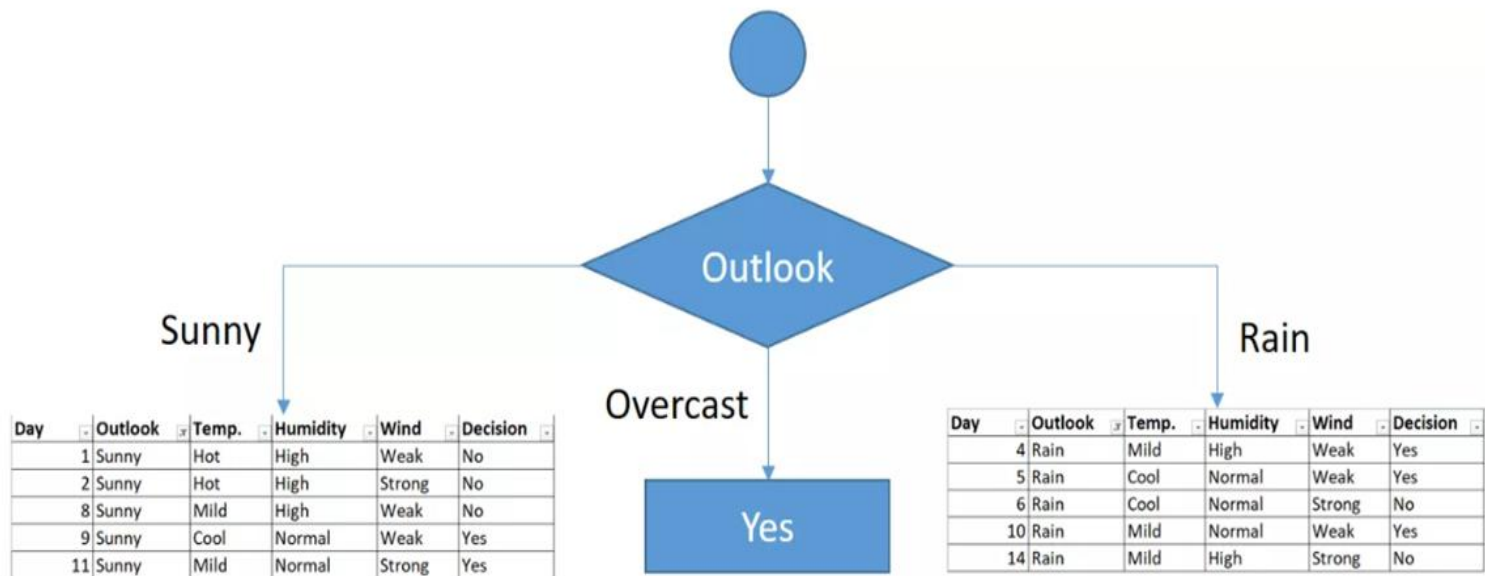


ID3算法

ID3 Algorithm

- Outlook作为决策树的根节点已位于树的最顶端，我们接下来重点关注它下面的三个分支叶子节点：晴天、多云、雨天（均为分类变量）。我们发现，多云（overcast）分支对应的子数据集中，结果全为“是（yes）”。这意味着多云这个分支已经处理完成，无需继续划分。

Now outlook is at the top of the tree. Then, we concentrate on leaves under: sunny, overcast, rain (Categorical variables). We also realize that sub dataset in the overcast leaf has only yes decisions. This means that overcast leaf is done.



Tree is over for overcast outlook leaf

ID3算法

ID3 Algorithm

- 在接下来的步骤中，我们将对这些子数据集采用相同的准则进行计算。例如，对outlook为晴天对应的子数据集，我们需要分别计算温度、湿度和风速这三个特征的信息增益。在outlook为晴天的条件下：总样本数为5，其中正例yes为2，反例no为3。

We will apply same principles to those sub datasets in the following steps. Focus on the sub dataset for sunny/outlook. We need to find the Gini index scores for temperature, humidity and wind features, respectively. Total population total under Outlook_sunny=5, yes=2, no=3.

Outlook = Sunny

Day	Temp.	Humidity	Wind	Decision
1	Hot	High	Weak	No
2	Hot	High	Strong	No
8	Mild	High	Weak	No
9	Cool	Normal	Weak	Yes
11	Mild	Normal	Strong	Yes

ID3算法

ID3 Algorithm

- 同理，我们需要在边为Sunny值情况下，决定下一节点的属性。根据上表，我们可以从温度、湿度、风力三个元素里面选。我们先计算各自的信息增益。

We first need to decide the attribute for the root node. According to the table above, we can choose from outlook, temperature, humidity and wind. We first calculate the information gain of outlook.

- 我们首先根据公式计算 $H(\mathcal{D})$ ，如下所示。同样的，我们简单地用频率计算先验概率：

First we calculate the parent entropy, aka., before classification using that attribute, as in the following. Similarly, we using the frequency of each case to approach the prior probability:

$$H(\mathcal{D}) = - \sum_{k=1}^K \frac{|C_k|}{|\mathcal{D}|} \log_2 \frac{|C_k|}{|\mathcal{D}|} = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

ID3算法

ID3 Algorithm

► 下面，我们计算关于温度的 $H(\mathcal{D}|A)$ ：

Now, we calculate the $H(\mathcal{D}|A)$ for temperature in the below:

Temperature	Yes	No	#instance
Hot	$0(P_{\text{yes}} = 0/2)$	$2(P_{\text{no}} = 2/2)$	2
Mild	$1(P_{\text{yes}} = 1/1)$	$0(P_{\text{no}} = 0/1)$	1
Cool	$1(P_{\text{yes}} = 1/2)$	$1(P_{\text{no}} = 1/2)$	2

$$\begin{aligned} H(\mathcal{D}|A) &= \sum_{i=1}^3 \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \sum_{k=1}^2 -\frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \\ &= \frac{2}{5} \left[-\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} \right] + \frac{1}{5} \left[-\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right] = 0.4 \end{aligned}$$

ID3算法

ID3 Algorithm

► 同理，关于湿度的 $H(\mathcal{D}|A)$ 计算如下：

Similarly, we calculate $H(\mathcal{D}|A)$ for humidity in the below:

Outlook	Yes	No	#instance
High	$0(P_{\text{yes}} = 0/3)$	$3(P_{\text{no}} = 3/3)$	3
Normal	$2(P_{\text{yes}} = 2/2)$	$0(P_{\text{no}} = 0/2)$	2

$$H(\mathcal{D}|A) = \sum_{i=1}^2 \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \sum_{k=1}^2 -\frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} = \frac{3}{5} \left[-\frac{0}{3} \log_2 \frac{0}{3} - \frac{3}{3} \log_2 \frac{3}{3} \right] + \frac{2}{5} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right]$$

= 0

ID3算法

ID3 Algorithm

► 同理，关于风力的 $H(\mathcal{D}|A)$ 计算如下：

Similarly, we calculate $H(\mathcal{D}|A)$ for wind in the below:

Temperature	Yes	No	#instance
Weak	$1(P_{\text{yes}} = 1/3)$	$2(P_{\text{no}} = 2/3)$	3
Strong	$1(P_{\text{yes}} = 1/2)$	$1(P_{\text{no}} = 1/2)$	2

$$H(\mathcal{D}|A) = \sum_{i=1}^2 \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \sum_{k=1}^2 -\frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} \log_2 \frac{|\mathcal{D}_{ik}|}{|\mathcal{D}_i|} = \frac{3}{5} \left[-\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right]$$
$$= 0.95$$

ID3算法

ID3 Algorithm

- 根据上面结果，我们计算相应的信息增量，如下表所示：

Now we calculate the information gain according to the table below:

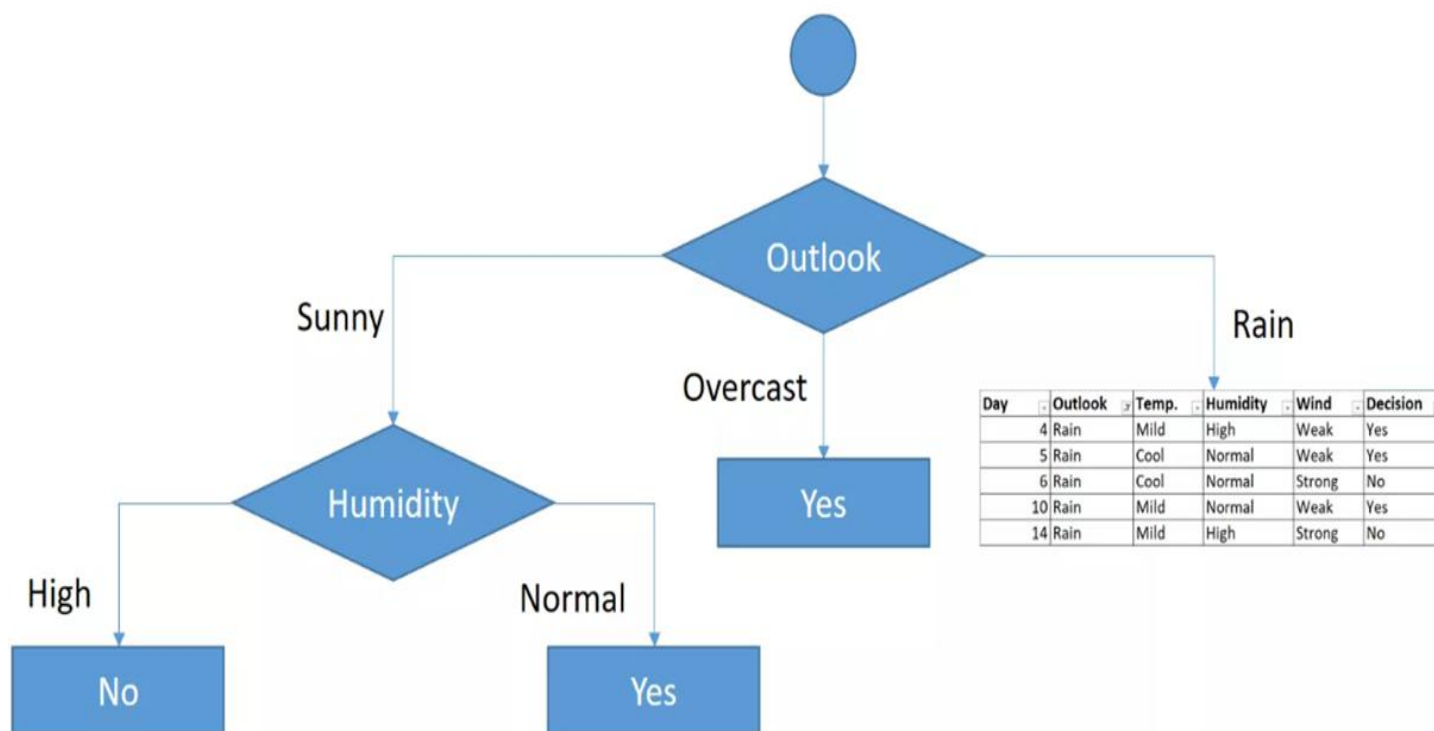
$H(\mathcal{D})$	$H(\mathcal{D} \text{Temp.})$	$H(\mathcal{D} \text{Humidity})$	$H(\mathcal{D} \text{Wind})$	Choice
0.97	0.4	0.0	0.95	
$g(\mathcal{D}, A)$	0.57	0.97	0.02	Humidity

- 从表中可以看出，根据上面结果，关于属性或特征Humidity的信息增量最大。我们选择属性Humidity作为节点，继续往下建树。所建的第二层级的树如下页所示：

From the table that, it is obvious that the attribute/feature “outlook” has the largest information gain based on the previous results. In the ID3 algorithm, the attribute with the largest information gain is selected as the node to grow the decision tree downward. The first-level tree constructed is shown on the next page:

ID3算法

ID3 Algorithm

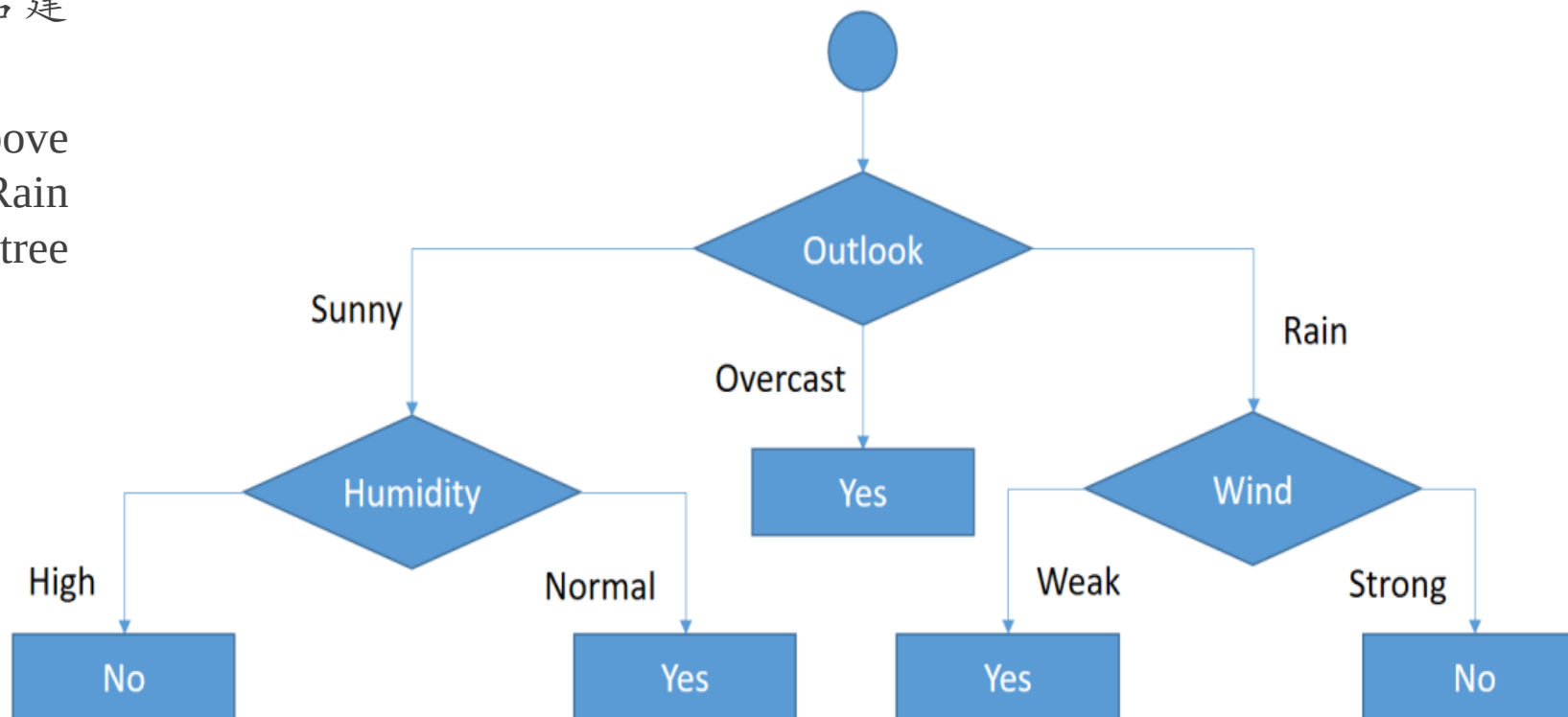


ID3算法

ID3 Algorithm

- 对 Rain 的边重复上面的工作，最后建的树如下所示：

Repeat the above calculation for Rain branch, the final tree is as in the right:



决策树

Decision Tree

- 建决策树的方法是多种多样的，除了上面的ID3算法之外，还有其它算法，例如C4.5、C5.0等，其基于不同的指标，如Gini指标，确定树的节点。

There are numerous implementations of decision trees, besides the ID3 algorithm detailed above, other well-known algorithms include the C4.5 algorithm, C5.0 algorithm, etc.

- 上们上面所讲的决策树是用于分类的例子，即将数据集划分为响应变量所属的不同类别，因此此类树称为分类树。除此之外，决策树还可以用于回归，即响应变量为数值型或连续型，此时决策树又称为回归树。特别的，通过二叉递归划分（每个节点只分成两个子分支）构建的树，既可以用于分类，又可以用于回归，称为分类与回归树。

The exemplified decision tree constructed above is for classification, aka., to separate the dataset into classes belonging to the response variable. Besides that, decision tree can also used for regression, when the response variable is numeric or continuous. Specifically, by using binary recursive partitioning (each node splits into exactly two branches), the constructed decision tree algorithm can both for classification and regression, it is called Classification And Regression Trees (CART).